

1 Introduction

Based on the target of the P30 adhesin, the bilayer structure of the airway surface liquid (ASL), and the fact that bacteria are fully immersed in the periciliary layer, the central question addressed by this model is: **How strong must P30-mediated adhesion be to achieve stable colonization in the dynamic fluid environment of the respiratory tract?**

Spatial geometry model

Bronchial epithelial cells: Adhesion substrate, expressing P30 receptors.

Periciliary layer (PCL): Thickness approximately $7\text{ }\mu\text{m}$, a low-viscosity fluid layer in which bacteria are completely immersed.

Mucus layer (ML): A highly viscoelastic fluid layer covering the PCL.

***E. coli* Nissle 1917:** Ellipsoidal cells with dimensions approximately $0.5 \times 2\text{ }\mu\text{m}$, expressing P30 adhesin on their surface.

Cilia: Length $5\text{ }\mu\text{m} \sim 7\text{ }\mu\text{m}$, generating periodic driving forces that propel fluid flow.

Core assumptions

Bacteria are fully immersed in the PCL, located between or above the cilia.

P30 directly binds to glycoprotein/glycolipid receptors on the epithelial cell surface (not to cilia).

Fluid shear forces arise primarily from cilium-driven flow within the PCL.

The mucus layer acts as an upper boundary and does not directly contact the bacteria.

2 Mechanical equilibrium equations

Force analysis on a bacterium

The force balance for an attached bacterium (steady state) is:

$$\sum \mathbf{F} = 0$$

where the dominant terms are the balance between **adhesion force** and **fluid drag force**.

The fluid drag force is described by a modified Stokes law accounting for wall effects:

$$F_{\text{drag}} = 6\pi \eta_{\text{PCL}} R v f_{\text{wall}}$$

η_{PCL} : PCL viscosity, approximately 7–10 times that of saline; R : equivalent hydrodynamic radius of the bacterium; v : local flow velocity generated by cilia; f_{wall} : wall correction factor (approximately 1.7 for a sphere near a surface).

The condition for stable adhesion is:

$$n \cdot f_b \geq F_{\text{drag}}$$

where n is the number of effective adhesive bonds and f_b is the single-bond strength.

Orientation effects

As a rod-shaped bacterium, its orientation significantly affects the drag force: in shear flow, the bacterium tends to reorient spontaneously with its long axis parallel to both the surface and the flow direction, thereby minimizing the force.

3 Adhesion bond dynamics

Effect of fluid drag on bond energy

The force-dependent dissociation rate is estimated using the Boltzmann distribution:

$$k_{\text{off}}(F) = k_{\text{off}}^0 \exp\left(\frac{F x_\beta}{k_B T}\right)$$

where: k_{off}^0 : dissociation rate at zero force; x_β : characteristic bond length (potential well width); k_B : Boltzmann constant.

Single-bond strength: Direct mechanical measurements for P30 are currently lacking; we refer to data for the FimH adhesin of type I pili in *E. coli*, whose single-bond strength is approximately 120 pN. The single-bond strength f_b of P30 is estimated to be in the range **100–150 pN** as a baseline for the model.

Dynamic equation for the number of adhesive bonds

The time evolution of the number of bonds n is described by:

$$\frac{dn}{dt} = k_{\text{on}} (N_{\text{max}} - n) - k_{\text{off}}(F) n$$

The steady-state solution is:

$$n_{\text{ss}} = \frac{N_{\text{max}}}{1 + \frac{k_{\text{off}}(F)}{k_{\text{on}}}}$$

where N_{max} is the maximum possible number of bonds, limited by: P30 expression level (to be determined experimentally); receptor density on the epithelial cell surface (to be obtained from literature); for a rod-shaped bacterium lying on its side, the contact area can be approximated as a curved segment along the long axis, with a typical contact area of about 0.1–0.5 μm^2 .

4 Dynamic force source: cilium-driven oscillatory flow

Velocity field model

Flow in the PCL is driven by ciliary beating, exhibiting periodicity and spatial inhomogeneity. It can be simplified as:

$$v(y, t) = v_{\text{max}}(y) [1 + \varepsilon \sin(\omega t + \phi)]$$

where: $v_{\text{max}}(y)$: depth-dependent velocity profile; ε : oscillation amplitude, approximately 0.3–0.5; ω : ciliary beating angular frequency, approximately

10–20 Hz; ϕ : phase. Boundary conditions: At the ciliary root ($y = 0$), $v = 0$ (no slip); At the ciliary tip ($y \approx 7 \mu\text{m}$), v reaches its maximum.

5 Summary of parameter sources

The parameters required for the model and their sources or estimation methods are as follows:

PCL viscosity η_{PCL} : Literature values, approximately 7–10 times the viscosity of saline.

Bacterial equivalent radius R : Geometric estimate, approximately $0.5 \mu\text{m}$.

Local flow velocity v_{avg} : Typical cilium-driven values, 20–100 $\mu\text{m/s}$.

P30 single-bond strength f_b : Referencing FimH data, estimated as 100–150 pN.

P30 expression level N_{P30} : To be determined experimentally.

Receptor density: To be obtained from literature or experiments.

Characteristic bond length x_β : 0.1–1 nm.

Zero-force dissociation rate k_{off}^0 : Literature analogy, 10^{-3} – 10^{-1}s^{-1} .

Association rate k_{on} : To be estimated from experimental data.

6 Estimation of results

Prediction of the minimum P30 surface density

Based on the force balance condition, we use the maximum velocity and the maximum fluid drag force over time, together with the minimum adhesion force, to estimate:

$$N_{\text{max}} = \frac{F_{\text{drag}}}{f_b} \cdot (1 + \varepsilon) \cdot \left\{ 1 + \frac{k_{\text{off}}^0}{k_{\text{on}}} \cdot \exp \left[\frac{x_\beta F_{\text{drag}}}{k_B T} (1 + \varepsilon) \right] \right\}$$

Therefore:

$$n_{\text{P30,min}} = \frac{N_{\text{max}}}{A_{\text{contact}}}$$

Numerical estimates yield a requirement of **approximately 10 P30 proteins per square micrometer**.

7 Advanced directions

If time and resources permit, the following aspects could be considered:

CFD simulation: Establish a CFD model with realistic airway geometry to compute the local shear stress distribution on the bacterial surface.

P30 expression: High-lactate environments may lead to downregulation of P30 expression.

Ciliary motion: Possible effects of ciliary motion on bond energy.

Collective effects: Fluid interactions during colonization by multiple bacteria.

8 References

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